EE4-10
EE9-CS5-1
EE9-SC3
EE9-AML2-01
EE9-FPN2-02

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Monday, 17 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Each of the four questions has 25 marks.

The Questions

1. Random variables.

a) Consider the following joint pdf's for the random variables X and Y, where c_1, c_2, c_3 are constants. Calculate c_1, c_2, c_3, derive the marginal pdf's of X and Y and determine whether X and Y are independent.

i) $f(x,y) = c_1 xy^2, \quad 0 < x < 1, \ 0 < y < 1.$ [5]

ii) $f(x,y) = c_2(xy^2 + x^2 y), \quad 0 < x < 1, \ 0 < y < 1.$ [5]

iii) $f(x,y) = c_3 e^{-x-y^2}, \quad 0 < x < \infty, \ -\infty < y < \infty.$ [5]

b) FOUR-envelope puzzle. There are four sealed envelopes, one of which containing a check of £1m. You bet one envelope contains the check. Then I open other two envelopes, showing there is no check inside. Find the a posteriori probabilities of the two unopened envelops, and determine whether you will stick to envelope you've chosen or switch to the other one.

[10]

2. Estimation and sequences of random variables.

a) The random variable X has the density of the Rayleigh distribution

$$f(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}, \qquad x > 0.$$

i) Derive the maximum-likelihood (ML) estimate of the parameter σ^2. [5]

ii) Find the ML estimate if we observe the i.i.d. samples x_i = 4, 3, 5, 7, 6. [2]

iii) Determine whether the ML estimate is unbiased. [3]

b) Let X be a Rayleigh random variable as defined above. Estimate the tail probability $P(X > a)$ where $a = 3\sigma$ using

i) Markov inequality; [5]

ii) Chebyshev inequality (assuming $a > 2E[X]$); [5]

iii) Exact cumulative distribution function. [5]

Hint: For the Rayleigh distribution, $E[X] = \sqrt{\frac{\pi}{2}}\sigma$, $Var[X] = \frac{4-\pi}{2}\sigma^2$.

3. Random processes.

a) Consider the random process $X(n) = A\cos(n\lambda) + B\sin(n\lambda)$ for parameter $\lambda > 0$, where A and B are correlated random variables with zero means and unit variances. Let $E[AB] = \frac{1}{2}$.

i) Find the mean and autocorrelation function of $X(n)$. [7]

ii) Is $\{X(n)\}$ stationary? Justify your answer. [3]

b) Consider the matched filter shown in the following figure:

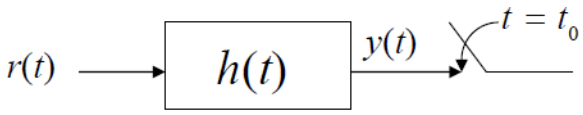


Fig. 3.1. Matched filter, which can be non-causal in this problem.

Let $r(t)$ represent a deterministic signal $s(t)$ corrupted by noise

$$r(t) = s(t) + w(t), \quad -\infty < t < \infty$$

where the signal

$$s(t) = \frac{1}{\sqrt{2\pi}\sigma_s} e^{-\frac{t^2}{2\sigma_s^2}}, \quad -\infty < t < \infty$$

and $w(t)$ is additive white noise with power spectral density

$$S_W(\omega) = N_0 e^{-\sigma_w^2 \omega^2/2}, \quad -\infty < \omega < \infty.$$

It is assumed that $\sigma_s^2 > \sigma_w^2$.

i) Derive the expression of the optimum matched filter $h(t)$. [10]

ii) Find the maximum output signal-to-noise ratio (SNR). [5]

4. Markov chains and martingales.

a) In an old house, white mice and black mice arrive as independent Poisson processes with intensity 0.1 and 0.2, i.e., at a rate of 1 white mouse/10 minutes and at a rate of 1 black mouse/5 minutes, respectively.

i) What is the probability that the first mouse is black? [5]

ii) What is the probability that the first mouse is black while the second is white? [5]

iii) Two cats are waiting in the house so that each mouse is caught randomly with equal probability. One cat catches a mouse every 10 minutes, and the other cat catches a mouse every 20 minutes. Find the average number of mice in the house. [5]

b) Consider a symmetric random walk on the integers $\{0, \pm 1, \pm 2, \dots\}$. This Markov chain is a sequence $\{X_n\}$ where $P\{X_{n+1} = X_n + 1\} = \frac{1}{3}, P\{X_{n+1} = X_n - 1\} = \frac{2}{3}$. Suppose the chain starts at the origin $X_0 = 0$.

i) Derive the probability $P\{X_{2n} = 0\}$. [4]

ii) Using the Stirling formula $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, determine whether the origin is a recurrent or transient state. [6]